微塑料：
无处不在，危害几何？

Microplastics: Everywhere, but How Harmful?



检出了微塑料，就能证明它有害吗？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In May 2004 Richard Thompson, a marine biologist at the University of Plymouth, and colleagues described an overlooked pollutant in the journal Science. Sifting through plankton samples kept since the 1960s, they found that microscopic plastic fragments and synthetic fibres had grown more abundant. They labelled the particles microplastics.

Since then the durable debris has proved ubiquitous, turning up on shorelines, in deep-sea sediment, in rainfall, and in Arctic ice. Regulation came before proof of harm to people: in December 2015 America passed a law banning plastic microbeads from scrubs and other cosmetics rinsed down drains.

The search turned inward. In March 2022 a small Dutch study found plastic in the blood of about three-quarters of 22 healthy donors. In March 2024 a study in the New England Journal of Medicine linked plastic in the fatty plaque of neck arteries to a higher risk of heart attack, stroke, or death. Its authors cautioned that correlation is not causation.

The most sensational figures are the shakiest. In June 2019 a report for WWF, a conservation group, reckoned people might swallow a credit card's worth of plastic a week; critics call that an overestimate. Two months later the World Health Organization was reassuring but tentative: on limited evidence, microplastics in drinking water posed no apparent risk at current levels. In February 2025 a study in Nature Medicine reported plastic in brain tissue in an amount popularised as a spoonful; others queried its methodology.

Detection has outrun understanding; human studies remain small or observational. Harm is plausible but unproven, and until the verdict is in, better measurement may guide policy more safely than panic.

中文译文

2004年5月，普利茅斯大学海洋生物学家理查德·汤普森和同事在《科学》杂志上描述了一种长期被忽略的污染物。他们细查自20世纪60年代起保存的浮游生物样本，发现其中微小的塑料碎片和合成纤维越来越多。他们把这些颗粒称为微塑料。

此后，人们发现，这些难以消失的碎屑几乎无处不在：海岸、深海沉积物、雨水和北极冰层中都有它们的踪迹。监管先于人体危害的确证出现：2015年12月，美国通过法律，禁止在磨砂膏等用后冲入下水道的化妆品中添加塑料微珠。

寻找微塑料的目光随后转向人体内部。2022年3月，一项小规模荷兰研究在22名健康献血者中，发现约四分之三的人血液里有塑料。2024年3月，《新英格兰医学杂志》发表的一项研究发现，颈部动脉脂肪性斑块中的塑料与较高的心脏病发作、中风或死亡风险相关。作者提醒：相关不等于因果。

最耸人听闻的数字，往往也最站不住脚。2019年6月，一份为自然保护组织WWF撰写的报告估计，人每周可能吞下相当于一张信用卡重量的塑料；批评者认为这个数字过高。两个月后，世界卫生组织给出了令人稍感安心、却仍属初步的判断：依据有限证据，饮用水中的微塑料在当时的含量下并无明显风险。2025年2月，《自然·医学》上的一项研究报告了脑组织中的塑料，其数量被大众转述成“一勺”；另一些人则质疑这项研究的方法。

检测走在了理解前面；人体研究仍然规模较小，或只属于观察性研究。微塑料可能有害，但尚未得到证实。在结论明确之前，用更准确的测量来指导政策，或许比任由恐慌主导更稳妥。

核心词汇

先记这22项。连着下方的原文短语一起回想。

pollutant n.
污染物
described an overlooked pollutant in

sift through phr.v.
细查；逐一筛查
Sifting through plankton samples kept

microscopic adj.
极微小的；显微镜下的
that microscopic plastic fragments and

synthetic adj.
合成的；人造的
plastic fragments and synthetic fibres

abundant adj.
大量存在的
fibres had grown more abundant.

durable adj.
耐久的；不易分解的
Since then the durable debris has

debris n.
碎屑；残片
Since then the durable debris has

ubiquitous adj.
无处不在的
durable debris has proved ubiquitous,

sediment n.
沉积物
on shorelines, in deep-sea sediment,

regulation n.
监管；规管
Regulation came before proof of harm

donor n.
献血者
three-quarters of 22 healthy donors.

artery n.
动脉
in the fatty plaque of neck arteries

correlation n.
相关性
Its authors cautioned that correlation

causation n.
因果关系
that correlation is not causation.

sensational adj.
耸人听闻的
The most sensational figures are the

overestimate n.
过高的估计
critics call that an overestimate.

tentative adj.
初步的；未定的
was reassuring but tentative:

methodology n.
研究方法；方法体系
others queried its methodology.

observational adj.
观察性的
studies remain small or observational.

plausible adj.
有可能的；合乎情理的
Harm is plausible but unproven,

verdict n.
定论；判断
and until the verdict is in, better

measurement n.
测量
is in, better measurement may guide

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

marine adj.
海洋的

biologist n.
生物学家

plankton n.
浮游生物

fragment n.
碎片

fibre n.
纤维

label v.
把……称为；命名

particle n.
颗粒；微粒

turn up phr.v.
出现；被发现

shoreline n.
海岸线

rainfall n.
降雨；雨水

pass a law v.phr.
通过一项法律

scrub n.
磨砂清洁产品

cosmetic n.
化妆品

rinse down phr.v.
随冲洗流入

turn inward v.phr.
转向内部；此处指人体内

fatty adj.
含脂肪的

plaque n.
血管内的斑块

stroke n.
中风

caution v.
告诫；提醒谨慎

shaky adj.
最不可靠的

conservation n.
自然保护

reckon v.
估计；估算

swallow v.
吞下

reassuring adj.
令人安心的

apparent adj.
明显的

tissue n.
生物组织

spoonful n.
一勺的量

query v.
质疑

detection n.
检测；发现

unproven adj.
尚未证实的

panic n.
恐慌

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 _____ plankton samples kept since the 1960s,

02 Its authors cautioned that _____ is not causation.

03 Two months later the World Health Organization was reassuring but _____:

04 Harm is _____ but unproven,

文章主线

- 发现 2004年，汤普森团队在存档海洋样本中发现微小塑料碎片和合成纤维增多，将其称为微塑料。
- 扩散 微塑料从深海到北极无处不在；美国2015年立法禁用冲洗类化妆品塑料微珠，当时人体危害尚未证实。
- 证据 22名健康献血者中，约四分之三的人血中检出塑料；另一研究发现颈部动脉斑块中的塑料与更高的心梗、中风或死亡风险相关，不能据此证明因果。
- 争议 每周吞下信用卡分量塑料的估算被批高估；脑内塑料量被转述成“一勺”，研究方法受质疑。
- 判断 世卫2019年据有限证据，初判饮水微塑料在当时含量下无明显风险；人体危害仍未定论。

参考答案　1 Sifting through 2 correlation 3 tentative 4 plausible